CORE PURE / FURTHER CALCULUS

Integration by Parts

Mark scheme 15 questions · 105 marks

Questions

Question	Marks by part	Marks	Page
Question 1		5	4
Question 2	(a) 2 (b) 4	6	5
Question 3		7	6
Question 4		5	7
Question 5	(a) 2 (b) 5	7	8
Question 6	(a) 5 (b) 2	7	9
Question 7	(a) 5 (b) 2	7	10
Question 8	(a) 3 (b) 5 (c) 1	9	11
Question 9	(a) 3 (b) 5 (c) 1	9	12
Question 10		6	13
Question 11		6	14
Question 12		6	15
Question 13	(a) 2 (b) 4	6	16
Question 14	(a) 2 (b) 5	7	17
Question 15	(i) 1 (ii) 3 (iii) 8	12	18
	Total	**105**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$I = \left[\frac{2}{3}x^{3/2}\ln\left(\frac{9}{x}\right)\right]_1^9$ $-\int_1^9 \frac{2}{3}x^{3/2}\left(-\frac{1}{x}\right)\mathrm{d}x$	**M1 A1**	**M1** Integrates $\sqrt{x}$ and differentiates the logarithm, reaching $px^{3/2}\ln\left(\frac{9}{x}\right) \pm \int qx^{1/2}\,\mathrm{d}x$, oe after splitting the logarithm. Condone coefficient slips. **A1** Correct, oe. Limits not needed.
	$\frac{2}{3}\int x^{1/2}\,\mathrm{d}x = \frac{4}{9}x^{3/2}$	**A1**	
	$I = \left(\frac{2}{3}(27)\ln 1 + \frac{4}{9}(27)\right)$ $-\left(\frac{2}{3}\ln 9 + \frac{4}{9}\right)$	**dM1**	Substitutes 9 and 1 into their integrated expression and subtracts, upper minus lower.
	$I = \frac{104}{9} - \frac{4}{3}\ln 3$	**A1**	cso, or $a = \frac{104}{9}$, $b = -\frac{4}{3}$ stated. Not for $\ln 9$ left in the answer.
	Total for Question 1	**5**	

Question	Scheme	Marks	Notes
2(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{1+\left(\frac{x-2}{x+2}\right)^2} \times \frac{(x+2)-(x-2)}{(x+2)^2}$	**M1**	Uses $\frac{1}{1+w^2}$ times an attempt at the derivative of $\frac{x-2}{x+2}$ by quotient or product rule, or differentiates $\tan y = \frac{x-2}{x+2}$ implicitly. Condone sign slips.
	$= \frac{2}{x^2+4}$	**A1**	Simplified. Not for an unsimplified form such as $\frac{4}{2x^2+8}$.
		[2]	
2(b)	$u = \arctan\left(\frac{x-2}{x+2}\right)$, $v = x$ $I = x\arctan\left(\frac{x-2}{x+2}\right) - \int \frac{2x}{x^2+4}\,\mathrm{d}x$	**M1 A1**	**M1** Uses parts with $v = x$, reaching $x\arctan\left(\frac{x-2}{x+2}\right) - \int x\times$ their (a) $\mathrm{d}x$, their (a) any derivative found there. Condone a sign slip. **A1** oe.
	$\int \frac{2x}{x^2+4}\,\mathrm{d}x = \ln(x^2+4)$	**M1**	Integrates their $\frac{kx}{x^2+4}$ to $\lambda\ln(x^2+4)$, by substitution or inspection.
	$I = x\arctan\left(\frac{x-2}{x+2}\right) - \ln(x^2+4) + c$	**A1**	oe, including $+c$.
		[4]	
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3	$\left[\frac{x^3}{3}\ln(1+x)\right]_0^2$ $-\int_0^2 \frac{x^3}{3(1+x)}\,\mathrm{d}x$	**M1 A1**	**M1** Uses parts with $u=\ln(1+x)$, reaching the form $kx^3\ln(1+x)-\int\frac{mx^3}{1+x}\,\mathrm{d}x$. Condone coefficient slips. **A1** Limits need not appear yet.
	$\frac{x^3}{1+x}=x^2-x+1-\frac{1}{1+x}$	**M1 A1**	**M1** Divides x^3 by $1+x$, or writes x^3 in powers of $1+x$, reaching a quadratic plus $\frac{k}{1+x}$. Condone coefficient slips.
	$\int\frac{x^3}{1+x}\,\mathrm{d}x=\frac{x^3}{3}-\frac{x^2}{2}+x-\ln(1+x)$	**A1**	With or without the factor $\frac{1}{3}$.
	$\frac{8}{3}\ln 3-\frac{1}{3}\left(\frac{8}{3}-2+2-\ln 3\right)-0$	**dM1**	Substitutes $x=2$ and $x=0$, or the limits after a substitution, into both their uv term and their integral, and subtracts.
	$=3\ln 3-\frac{8}{9}$, so $a=-\frac{8}{9}$, $b=3$	**A1**	cso. $3\ln 3-\frac{8}{9}$ oe, or a and b stated; not $\ln 27-\frac{8}{9}$.
	Total for Question 3	**7**	

Question	Scheme	Marks	Notes
4	$t = 2x+1$: $\int \operatorname{arsinh}(2x+1)\,\mathrm{d}x = \frac{1}{2}\int \operatorname{arsinh} t\,\mathrm{d}t$ $= \frac{1}{2}\left(t\operatorname{arsinh} t - \int \frac{t}{\sqrt{t^2+1}}\,\mathrm{d}t\right)$	**M1 A1**	**M1** Uses parts with $u = \operatorname{arsinh}$ and $v = t$, x or $x+\frac{1}{2}$, differentiating arsinh to $\frac{k}{\sqrt{(2x+1)^2+1}}$ or $\frac{k}{\sqrt{t^2+1}}$. Condone slips in the factor $\frac{1}{2}$ or 2. **A1** Correct in either variable, such as $x\operatorname{arsinh}(2x+1) - \int \frac{2x}{\sqrt{(2x+1)^2+1}}\,\mathrm{d}x$.
	$\int \frac{t}{\sqrt{t^2+1}}\,\mathrm{d}t = \sqrt{t^2+1}$	**M1**	Integrates their remaining integral to $\lambda\sqrt{t^2+1}$, or in x to a form including $\lambda\sqrt{(2x+1)^2+1}$, by inspection or substitution.
	$\frac{1}{2}(2x+1)\operatorname{arsinh}(2x+1)$ $-\frac{1}{2}\sqrt{(2x+1)^2+1}$	**A1**	oe in x; $+c$ not needed here.
	$\frac{1}{2}(2x+1)\ln\left(2x+1+\sqrt{(2x+1)^2+1}\right)$ $-\frac{1}{2}\sqrt{(2x+1)^2+1}+c$	**A1**	oe, with $+c$. A0 if any arsinh remains.
	Total for Question 4	**5**	

Question	Scheme	Marks	Notes
5(a)	$h(\theta) = 3\sin^2\theta + 5\sin\theta\cos\theta - 2\cos^2\theta$ $= \frac{3}{2}(1-\cos 2\theta) + \frac{5}{2}\sin 2\theta - (1+\cos 2\theta)$	**M1**	Expands and replaces $\sin^2\theta$, $\cos^2\theta$ and $\sin\theta\cos\theta$ by double-angle forms, or expands the printed form in θ to compare. Condone sign slips.
	$h(\theta) = \frac{1}{2} - \frac{5}{2}\cos 2\theta + \frac{5}{2}\sin 2\theta$	**A1 AG**	Correct expansion and identities leading without error to the printed result; if both sides are worked, a conclusion is needed.
		[2]	
5(b)	$\int \left(\frac{\pi}{2} - \theta\right)\cos 2\theta \, \mathrm{d}\theta$ $= \frac{1}{2}\left(\frac{\pi}{2} - \theta\right)\sin 2\theta + \frac{1}{2}\int \sin 2\theta \, \mathrm{d}\theta$	**M1 A1**	**M1** Uses (a) and integrates by parts, differentiating the linear factor, on a product with $\cos 2\theta$ or $\sin 2\theta$, reaching the form $uv - \int v\frac{\mathrm{d}u}{\mathrm{d}\theta}\,\mathrm{d}\theta$. Condone sign and coefficient slips. **A1** One by-parts result correct; need not be simplified.
	$I = \left[\frac{\pi\theta}{4} - \frac{\theta^2}{4}\right.$ $- \frac{5}{4}\left(\frac{\pi}{2} - \theta\right)\sin 2\theta + \frac{5}{8}\cos 2\theta$ $\left. - \frac{5}{4}\left(\frac{\pi}{2} - \theta\right)\cos 2\theta - \frac{5}{8}\sin 2\theta\right]_0^{\frac{\pi}{2}}$	**A1**	Every term integrated correctly, combined or as separate integrals; need not be simplified.
	$= \left(\frac{\pi^2}{16} - \frac{5}{8}\right) - \left(\frac{5}{8} - \frac{5\pi}{8}\right)$	**dM1**	Substitutes $\frac{\pi}{2}$ and 0 into their integrated terms and subtracts, upper minus lower.
	$= \frac{\pi^2+10\pi-20}{16}$	**A1**	oe exact, such as $\frac{\pi^2}{16} + \frac{5\pi}{8} - \frac{5}{4}$.
		[5]	
	Total for Question 5	**7**	

Question	Scheme	Marks	Notes
6(a)	$\int x^2 \arctan x \, \mathrm{d}x$ $= \frac{x^3}{3} \arctan x - \int \frac{x^3}{3(1+x^2)} \, \mathrm{d}x$	**M1 A1**	**M1** Applies parts with $u = \arctan x$, reaching the form $px^3 \arctan x - q\int \frac{x^3}{1+x^2} \, \mathrm{d}x$. Condone sign and coefficient slips. **A1** Correct, oe.
	$\frac{x^3}{1+x^2} = x - \frac{x}{1+x^2}$	**M1**	Divides to the form $Ax + \frac{Bx}{1+x^2}$, or uses an equivalent substitution such as $t = 1 + x^2$. Condone coefficient slips.
	$\int \left(x - \frac{x}{1+x^2}\right) \mathrm{d}x = \frac{x^2}{2} - \frac{1}{2}\ln(1+x^2)$	**A1**	Both terms correct, oe, including in a substitution variable.
	$\frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{1}{6}\ln(1+x^2) + c$	**A1**	oe, with $+c$.
		[5]	
6(b)	$\left(\frac{1}{3}\arctan 1 - \frac{1}{6} + \frac{1}{6}\ln 2\right) - \left(\frac{1}{6}\ln 1\right)$	**M1**	Substitutes $x = 1$ and $x = 0$ into their answer to (a), containing an $\arctan x$ term, and subtracts. Zero lower-limit terms need not be shown.
	$\frac{\pi}{12} - \frac{1}{6} + \frac{\ln 2}{6}$ $\left[= \frac{\pi + 2\ln 2 - 2}{12}\right]$	**A1**	Any exact equivalent.
		[2]	
	Total for Question 6	**7**	

Question	Scheme	Marks	Notes
7(a)	$\int x^3 \arctan x \, \mathrm{d}x$ $= \frac{x^4}{4} \arctan x - \int \frac{x^4}{4(1+x^2)} \, \mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = \arctan x$, reaching the form $px^4 \arctan x - q\int \frac{x^4}{1+x^2} \, \mathrm{d}x$. Condone coefficient slips.
	$\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}$	**M1**	Divides, or splits $x^4 = (x^4 - 1) + 1$, reaching the form $ax^2 + b + \frac{c}{1+x^2}$. Condone coefficient slips.
	$\frac{1}{4}\int \left(x^2 - 1 + \frac{1}{1+x^2}\right) \mathrm{d}x$ $= \frac{1}{4}\left(\frac{x^3}{3} - x + \arctan x\right)$	**A1**	Correct integral, with or without the $\frac{1}{4}$.
	$\frac{x^4-1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4} + c$	**A1**	Any equivalent form, with $+c$.
		[5]	
7(b)	$\left[\text{“}\frac{x^4-1}{4} \arctan x - \frac{x^3}{12} + \frac{x}{4}\text{”}\right]_0^{\sqrt{3}}$ $= \frac{8}{4} \cdot \frac{\pi}{3} - \frac{3\sqrt{3}}{12} + \frac{\sqrt{3}}{4} - 0$	**M1**	Substitutes $\sqrt{3}$ and 0 into their answer to (a), which has an $\arctan x$ term, and subtracts. A zero lower value may be implied.
	$\frac{2\pi}{3}$	**A1**	cso, from a correct antiderivative.
		[2]	
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8(a)	$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{2}\ln(1+x)-\frac{1}{2}\ln(1-x)\right]$ $=\frac{1}{2}\left[\frac{1}{1+x}+\frac{1}{1-x}\right]$	**M1 A1**	**M1** Differentiates $\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$, split into two logarithms or by the chain rule. Condone sign and coefficient slips. Not for differentiating $\tanh y = x$. **A1** Any correct unsimplified form.
	$=\frac{1}{2}\cdot\frac{(1-x)+(1+x)}{(1+x)(1-x)}=\frac{1}{1-x^2}$	**A1 AG**	Correct derivative leading without error to $\frac{1}{1-x^2}$; combined steps accepted.
		[3]	
8(b)	$\int \operatorname{artanh} x\,\mathrm{d}x$ $= x\operatorname{artanh} x-\int\frac{x}{1-x^2}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $v = x$ and their derivative from (a), reaching $x\operatorname{artanh} x \pm \int x\,f(x)\,\mathrm{d}x$.
	$= x\operatorname{artanh} x+\frac{1}{2}\ln(1-x^2)$	**A1**	oe, such as $+\frac{1}{2}\ln(1-x)+\frac{1}{2}\ln(1+x)$ by partial fractions.
	$\left[\text{“}x\operatorname{artanh} x+\frac{1}{2}\ln(1-x^2)\text{”}\right]_0^{1/2}$ $=\frac{1}{2}\operatorname{artanh}\frac{1}{2}+\frac{1}{2}\ln\frac{3}{4}-0$ $\operatorname{artanh}\frac{1}{2}=\frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)\ [=\frac{1}{2}\ln 3]$	**dM1**	Substitutes both limits into their integral of the form $x\operatorname{artanh} x + k\ln(1-x^2)$ and writes $\operatorname{artanh}\frac{1}{2}$ as a logarithm.
	$\frac{3}{4}\ln 3-\ln 2$	**A1**	Any exact logarithmic form, such as $\frac{1}{4}\ln\frac{27}{16}$, with no artanh remaining.
		[5]	
8(c)	$\operatorname{artanh} x$ is defined only for $\lvert x\rvert<1$ $0\le x\le\frac{3}{2}$ includes $x=1$ and $x>1$	**E1**	States that $\operatorname{artanh} x$ is undefined, or not real, for $x \ge 1$ (or $x > 1$) and that the interval contains such values.
		[1]	
	Total for Question 8	**9**	

Question	Scheme	Marks	Notes
9(a)	$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\left[\ln(1+x)-\ln(1-x)\right]\right)$ $=\frac{1}{2}\left(\frac{1}{1+x}+\frac{1}{1-x}\right)$	**M1 A1**	**M1** Differentiates the logarithmic form, by splitting the logarithm or by the chain or quotient rule, to terms in $\frac{1}{1+x}$ and $\frac{1}{1-x}$. Condone sign slips. **A1** Correct derivative, unsimplified.
	$=\frac{1}{2}\left(\frac{(1-x)+(1+x)}{1-x^2}\right)=\frac{1}{1-x^2}$	**A1 AG**	Combines over a common denominator to the printed result without error.
		[3]	
9(b)	$\int_0^{1/3}\operatorname{artanh}(2x)\,\mathrm{d}x$ $=\left[x\operatorname{artanh}(2x)\right]_0^{1/3}-\int_0^{1/3}\frac{2x}{1-4x^2}\,\mathrm{d}x$	**M1 A1**	**M1** Uses parts with $v=x$ and a derivative of $\operatorname{artanh}(2x)$ of the form $\frac{k}{1-4x^2}$ from (a), reaching $x\operatorname{artanh}(2x)-\int\frac{kx}{1-4x^2}\,\mathrm{d}x$.
	$\int\frac{2x}{1-4x^2}\,\mathrm{d}x=-\frac{1}{4}\ln(1-4x^2)$	**A1**	oe, including by substitution with correct limits in the new variable.
	$\frac{1}{3}\operatorname{artanh}\left(\frac{2}{3}\right)+\frac{1}{4}\ln\left(\frac{5}{9}\right)$ $\operatorname{artanh}\left(\frac{2}{3}\right)=\frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right)=\ldots$	**dM1**	Applies both limits to their expression and uses the logarithmic form to write $\operatorname{artanh}\left(\frac{2}{3}\right)$ as a logarithm.
	$\frac{5}{12}\ln 5-\frac{1}{2}\ln 3$	**A1**	oe exact logarithmic form, such as $\frac{1}{12}\ln\left(\frac{3125}{729}\right)$; a decimal scores A0. cso.
		[5]	
9(c)	At $x=\frac{1}{2}$, $2x=1$ and $\operatorname{artanh}1$ is undefined	**E1**	States the integrand is undefined or infinite at the upper limit, $x=\frac{1}{2}$, as $2x=1$. Not for a general domain remark without the limit.
		[1]	
	Total for Question 9	**9**	

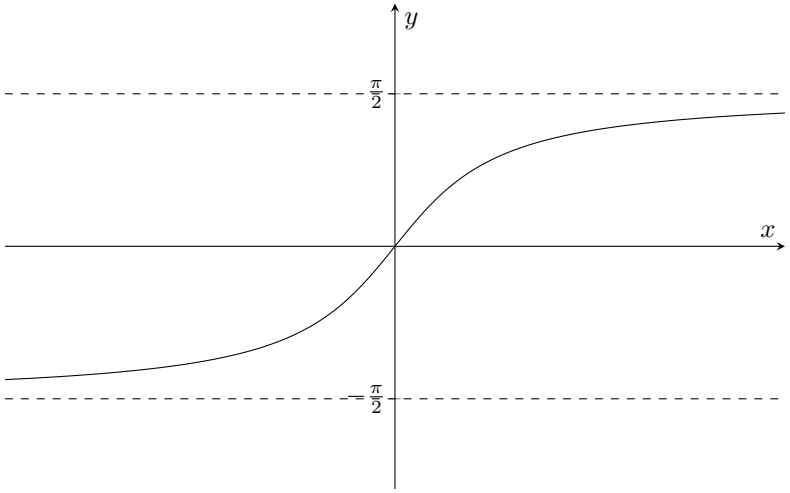
Question	Scheme	Marks	Notes
10	$I = \frac{x^2}{2}\operatorname{artanh}(2x-1)$ $\quad - \int \frac{x^2}{2} \cdot \frac{2}{1-(2x-1)^2}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = \operatorname{artanh}(2x-1)$, reaching $px^2\operatorname{artanh}(2x-1) - \int \frac{qx^2}{1-(2x-1)^2}\,\mathrm{d}x$. Condone coefficient slips. **A1** Correct expression; need not be simplified.
	$\frac{x}{1-x} = -1 + \frac{1}{1-x}$ $\Rightarrow \int \frac{x}{1-x}\,\mathrm{d}x = \ldots$	**M1**	Writes their rational integrand as a constant plus $\frac{k}{1-x}$, by division or substitution, and integrates to the form $ax + b\ln(1-x)$.
	$I = \frac{x^2}{2}\operatorname{artanh}(2x-1) + \frac{x}{4} + \frac{1}{4}\ln(1-x)$	**A1**	oe; c not needed here.
	$\operatorname{artanh}(2x-1) = \frac{1}{2}\ln\left(\frac{1+(2x-1)}{1-(2x-1)}\right)$ $= \ldots \ [= \frac{1}{2}\ln\frac{x}{1-x}]$	**M1**	Uses $\operatorname{artanh} t = \frac{1}{2}\ln\frac{1+t}{1-t}$ with $t = 2x-1$ in their expression. Condone slips simplifying the argument.
	$I = \frac{x^2}{4}\ln x + \frac{1-x^2}{4}\ln(1-x) + \frac{x}{4} + c$	**A1**	Any correct form in logarithms with no artanh, such as $\frac{x^2}{4}\ln\frac{x}{1-x} + \frac{1}{4}\ln(1-x) + \frac{x}{4} + c$. Needs $+c$.
	Total for Question 10	**6**	

Question	Scheme	Marks	Notes
11	$\int_0^3 (x^3 - 6x^2 + 9x)e^{-x}\,\mathrm{d}x$ $= -(x^3 - 6x^2 + 9x)e^{-x}$ $\quad + \int (3x^2 - 12x + 9)e^{-x}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with the polynomial, expanded or not, differentiated, reaching $\pm p(x)e^{-x} \pm \int q(x)e^{-x}\,\mathrm{d}x$. Condone sign slips. **A1** Correct first application, oe.
	$= -(x^3 - 6x^2 + 9x)e^{-x} - (3x^2 - 12x + 9)e^{-x}$ $\quad - (6x - 12)e^{-x} - 6e^{-x}$	**dM1**	Applies parts again, or a tabular method, until no integral remains. Condone sign and coefficient slips.
	$= (-x^3 + 3x^2 - 3x - 3)e^{-x}$	**A1**	Correct antiderivative, oe; need not be simplified.
	$[\ldots]_0^3 = (-27 + 27 - 9 - 3)e^{-3} - (-3)e^0$	**dM1**	Substitutes $x = 3$ and $x = 0$ into their antiderivative of the form $p(x)e^{-x}$ and subtracts; the lower limit is not taken as zero.
	$3 - 12e^{-3}$	**A1**	cso. In the form $A + Be^{-3}$, or $A = 3$, $B = -12$; $3 - \frac{12}{e^3}$ scores A0.
	Total for Question 11	**6**	

Question	Scheme	Marks	Notes
12	$t = \ln x \Rightarrow \mathrm{d}x = e^t\,\mathrm{d}t$ $x = e^{-1/2} \Rightarrow t = -\frac{1}{2}$, $x = 1 \Rightarrow t = 0$	**M1**	Substitutes $t = \ln x$ or $x = e^t$, replacing $\mathrm{d}x$ by $e^t\,\mathrm{d}t$ oe, and changes the limits or later returns to x.
	$I = \int_{-1/2}^{0} t^2 e^{2t}\,\mathrm{d}t$	**A1**	
	$\int t^2 e^{2t}\,\mathrm{d}t = \frac{1}{2}t^2e^{2t} - \int te^{2t}\,\mathrm{d}t$ $= \frac{1}{2}t^2e^{2t} - \frac{1}{2}te^{2t} + \int \frac{1}{2}e^{2t}\,\mathrm{d}t$	**M1**	Applies integration by parts twice, each time to the form $pt^ke^{2t} - \int qt^{k-1}e^{2t}\,\mathrm{d}t$. Condone sign and coefficient slips.
	$e^{2t}\left(\frac{1}{2}t^2 - \frac{1}{2}t + \frac{1}{4}\right)$	**A1**	Correct antiderivative, oe; need not be simplified.
	$\frac{1}{4} - e^{-1}\left(\frac{1}{8} + \frac{1}{4} + \frac{1}{4}\right)$	**dM1**	Substitutes both limits into their antiderivative, in t or back in x, and subtracts the right way round.
	$\frac{1}{4} - \frac{5}{8e}$, so $a = \frac{1}{4}$, $b = -\frac{5}{8}$	**A1**	cso. Exact values; a decimal value of the integral scores A0.
Total for Question 12		**6**	

Question	Scheme	Marks	Notes
13(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-\left(\frac{x-1}{3}\right)^2}} \times \frac{1}{3}$	**M1**	Differentiates to the form $\frac{k}{\sqrt{1-\left(\frac{x-1}{3}\right)^2}}$, or $\frac{k}{\cos y}$ from $\sin y = \frac{x-1}{3}$. Condone a wrong or missing k.
	$= \frac{1}{\sqrt{9-(x-1)^2}}$	**A1**	Simplified, or $\frac{1}{\sqrt{8+2x-x^2}}$.
		[2]	
13(b)	$x \arcsin\left(\frac{x-1}{3}\right) - \int \frac{x}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x$	**M1 A1**	**M1** Applies parts with $v = x$ or $x-1$, reaching $v\arcsin\left(\frac{x-1}{3}\right) \pm \int v \times$ their (a) $\mathrm{d}x$, their (a) of the form $\frac{k}{\sqrt{9-(x-1)^2}}$. **A1** Correct, oe such as with $v = x-1$.
	$\int \frac{(x-1)+1}{\sqrt{9-(x-1)^2}}\,\mathrm{d}x$ $= -\sqrt{9-(x-1)^2} + \arcsin\left(\frac{x-1}{3}\right)$	**M1**	Integrates their remaining integral to the form $p\sqrt{9-(x-1)^2} + q\arcsin\left(\frac{x-1}{3}\right)$, by splitting the numerator or by substitution.
	$(x-1)\arcsin\left(\frac{x-1}{3}\right) + \sqrt{9-(x-1)^2} + c$	**A1**	oe, such as uncollected arcsin terms or $\sqrt{8+2x-x^2}$. Needs $+c$.
		[4]	
	Total for Question 13	**6**	

Question	Scheme	Marks	Notes
14(a)	$g(\theta) = 1 + \sin^2\theta + \cos^2\theta$ $\quad + 2\sin\theta - 2\cos\theta - 2\sin\theta\cos\theta$ $\quad = 1 + 1 + 2\sin\theta - 2\cos\theta - \sin 2\theta$	**M1**	Expands to six terms and uses both $\sin^2\theta + \cos^2\theta = 1$ and $2\sin\theta\cos\theta = \sin 2\theta$. Condone one sign slip in the expansion.
	$g(\theta) = 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta$	**A1 AG**	Correct expansion leading without error to the printed result.
		[2]	
14(b)	$\int_0^{\pi/2} \theta\,(2 + 2\sin\theta - 2\cos\theta - \sin 2\theta)\,\mathrm{d}\theta$ $\int \theta\sin\theta\,\mathrm{d}\theta = -\theta\cos\theta + \int \cos\theta\,\mathrm{d}\theta$	**M1**	Uses (a) and applies integration by parts to a term $\theta\sin k\theta$ or $\theta\cos k\theta$, differentiating θ. Condone sign and coefficient slips.
	$\int \theta\sin\theta\,\mathrm{d}\theta = -\theta\cos\theta + \sin\theta$	**A1**	Any one of the three products integrated correctly.
	$\int \theta\cos\theta\,\mathrm{d}\theta = \theta\sin\theta + \cos\theta$ $\int \theta\sin 2\theta\,\mathrm{d}\theta = -\frac{1}{2}\theta\cos 2\theta + \frac{1}{4}\sin 2\theta$ $\int 2\theta\,\mathrm{d}\theta = \theta^2$	**A1**	All four terms correct, combined or integrated separately.
	$\left(\frac{\pi^2}{4} + 2 - \pi - \frac{\pi}{4}\right) - (-2)$	**dM1**	Substitutes $\frac{\pi}{2}$ and 0 into their integrated terms and subtracts.
	$\frac{\pi^2 - 5\pi + 16}{4}$	**A1**	Any exact equivalent, such as $\frac{\pi^2}{4} - \frac{5\pi}{4} + 4$.
		[5]	
	Total for Question 14	**7**	

Question	Scheme	Marks	Notes
15(i)			
	Asymptotes	**B1**	Increasing curve through the origin, approaching both $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$, with both asymptotes labelled.
		[1]	
15(ii)	$y = \arctan x \Rightarrow x = \tan y$ $1 = \sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x}$	**M1**	Writes $x = \tan y$ and differentiates implicitly, or finds $\frac{\mathrm{d}x}{\mathrm{d}y} = \sec^2 y$ and inverts. Condone a slip in the derivative of $\tan y$.
	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y}$	**M1**	Uses $\sec^2 y = 1 + \tan^2 y$ to write their derivative in terms of x.
	$= \frac{1}{1+x^2}$	**A1 AG**	Correct working from $x = \tan y$, leading without error to the printed result in x.
		[3]	
15(iii)	$I = \left[\frac{x^2}{2}(\arctan x)^2\right]_0^1$ $- \int_0^1 \frac{x^2}{2} \cdot \frac{2\arctan x}{1+x^2}\,\mathrm{d}x$	**M1**	Integrates by parts with $u = (\arctan x)^2$, reaching $\frac{x^2}{2}(\arctan x)^2 - \int k\frac{x^2 \arctan x}{1+x^2}\,\mathrm{d}x$. Condone a slip in k.
	$= \frac{\pi^2}{32} - \int_0^1 \frac{x^2 \arctan x}{1+x^2}\,\mathrm{d}x$	**A1**	
	$\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$ $\Rightarrow I = \frac{\pi^2}{32} - \int_0^1 \arctan x\,\mathrm{d}x$ $+ \int_0^1 \frac{\arctan x}{1+x^2}\,\mathrm{d}x$	**M1**	Writes $\frac{x^2}{1+x^2}$ as $1 - \frac{1}{1+x^2}$, by division or otherwise, and splits their integral into two.
	$\int_0^1 \arctan x\,\mathrm{d}x = [x\arctan x]_0^1$ $- \int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x$	**M1**	Integrates by parts with $u = \arctan x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, reaching $x\arctan x \pm \int \frac{kx}{1+x^2}\,\mathrm{d}x$.
	$\int_0^1 \arctan x\,\mathrm{d}x = \frac{\pi}{4} - \frac{1}{2}\ln 2$	**A1**	$\int \frac{x}{1+x^2}\,\mathrm{d}x$ by substitution or by inspection.

Question 15 continued

Question	Scheme	Marks	Notes
	$t = \arctan x \Rightarrow \int_0^1 \frac{\arctan x}{1+x^2}\,\mathrm{d}x$ $= \int_0^{\pi/4} t\,\mathrm{d}t$	**M1**	Substitutes $t = \arctan x$ or $x = \tan t$, replacing $\mathrm{d}x$, to reach $\int t\,\mathrm{d}t$; or integrates by inspection to $k(\arctan x)^2$.
	$= \left[\frac{t^2}{2}\right]_0^{\pi/4} = \frac{\pi^2}{32}$	**A1**	
	$I = \frac{\pi^2}{32} - \left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right) + \frac{\pi^2}{32}$ $= \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2$	**A1 AG**	Correct values of all three terms combined with correct signs, leading without error to the printed result.
		[8]	
	Total for Question 15	**12**	